SVD Energy Day-Ahead-Prices

Denis Šmerda

Introduction

Hedging energy assets is an important aspect when trying to stabilize revenues. Because a forward curve consists of dozens of individual contracts, it is incredibly inefficient to try and hedge each maturity date separately. Instead, I need to identify the economic shocks that explain the volatility of these price changes. In this project, I apply SVD to historical market data to extract these hidden risk factors, while taking a closer look at the problem from different perspectives. Using this simplified market structure, I then implement least squares optimization to calculate the combination of exchange contracts needed to hedge an asset's risk.

1 Data

The dataset I am working with provides hourly day-ahead electricity prices in EUR/MWh for six western European countries. The data is relatively clean, there are only a few observations missing in each country. Moving forward, France is the country that will be analysed in this project, as the dataset is primarily focused on this country. Also, there is strong seasonality in daily energy prices, but it differs depending on the weekday. To make the models more accurate, I will focus only on business days from now on, but the models could also be applied to only weekend data.

To address the missing values problem, simple linear interpolation across hours is used, rather than dropping the affected rows or applying more complex techniques such as splines. After this change, the dataset has 24 columns denoting all hours of a day, and 260 observations which stand for individual business days ranging from 01/01/2022 to 31/12/2022. Further exploring the data, I continue with summary statistics of individual hours over the year, presented in Table 1.

Table 1: Summary statistics of individual hours

	3:00	9:00	15:00	21:00
mean	233.8	360.3	291.2	346.0
std	110.3	223.7	149.2	162.0
min	-0.0	22.0	28.6	15.0
25%	169.0	242.9	189.5	234.5
50%	203.8	304.3	237.2	290.6
75%	301.2	461.1	392.2	443.3
max	585.0	2987.8	810.7	870.0

Important discoveries of the summary statistics include the fact that prices reach negative values and there are some rather high observations. The unusually high prices were recorded on the fourth of April from 8:00 to 10:00. Various sources state that this was a supply shortage at a time of huge demand, naturally increasing the price, thus it is not an observation error. As this is a tail event, unlikely to repeat consistently, the two outlying values are removed and then linearly interpolated along the day observations.

In Figure 1 I plot the midday prices over the whole year. They are certainly

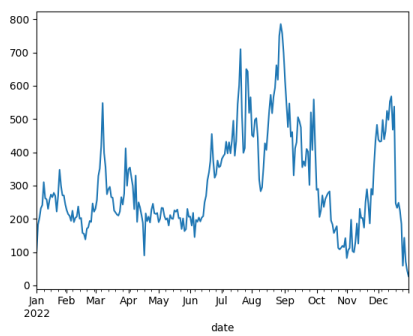


Figure 1: Prices at Midday

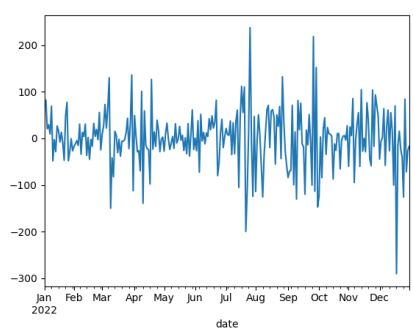


Figure 2: Midday price differences

very volatile and the volatility does not seem to be constant. After running an ADF test on every column of this dataset, at the 5% significance level, there is not enough evidence to reject the null hypothesis of a unit root for 23 of them. The following formula

$$\Delta P_{day,hour} = P_{day,hour} - P_{day-1,hour}$$

is applied to the whole dataset, resulting in Figure 2. It transforms the day-ahead prices to day-ahead price differences and is introduced to make all of the column time series stationary. It has to transform the whole dataset, otherwise the SVD results would not make sense. With this action the dataset is losing its first observation. Logarithmic differences were not available in this dataset because of negative values.

A plot of differences in prices of each hour over the year can be seen in Figure 3. To deal with the seasonality issue, the following within transformation

$$y^{centered}_{day,hour} = \Delta P_{day,hour} - \frac{1}{N}\sum_{day=1}^{N} \Delta P_{day,hour}$$

is applied. Midday absolute demeaned differences in day-ahead prices throughout

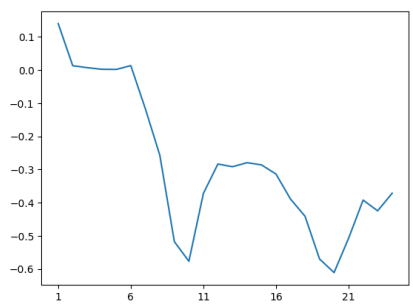


Figure 3: Hour averages

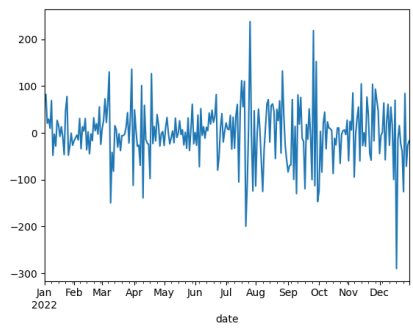


Figure 4: Demeaned midday price differences

the year can be seen in Figure 4.

A portion of the summary statistics table after all preceding transformations is presented as Table 2.

Table 2: Summary statistics of individual hours after transformation

	3:00	9:00	15:00	21:00
mean	-0.0	0.0	0.0	0.0
std	50.43	67.59	61.63	54.44
min	-194.81	-292.64	-337.42	-191.00
25%	-18.22	-28.95	-27.91	-24.71
50%	1.06	0.60	2.17	-0.35
75%	23.45	30.14	30.85	21.99
max	248.06	266.06	268.48	249.33

2 SVD application

The previous section led to the application of the singular value decomposition, which is used in the majority of data-related fields. In this study, I would like to take a look at interpretations from four points of view and present them one by one. I will begin with general math, more precisely linear algebra, which explains how the model works and how it can be applied to my dataset. Next,

I will translate these mathematical components into practical energy market concepts and check the numerical properties of the methods. Lastly, to bring deeper insight into this topic, I will move on to the economic aspect of energy prices with supply and demand tools.

2.1 Linear Algebra

Let me suppose that $A = (a_{i,j})_{m,n}$, where $\forall i \in \{1, \ldots, m\}, j \in \{1, \ldots, n\} : a_{i,j} \in \mathbb{R}$ and its rank is $r \in \mathbb{N}$. Further, $\mathbb{R}^m$ is a vector space and $\forall j : \vec{a}_j \in \mathbb{R}^m$. With these assumptions, there exist orthonormal bases $B = (\vec{v}_1, \ldots, \vec{v}_n)$, $C = (\vec{u}_1, \ldots, \vec{u}_m)$ of vector spaces $\mathbb{R}^n, \mathbb{R}^m$ respectively and real values $\sigma_1, \ldots, \sigma_r > 0$. This allows for the following decomposition

$$A = UDV^{-1} = UDV^T,$$

where $U = [id]^C_{K_m} = (\vec{u}_1, \ldots, \vec{u}_m)$, $V = [id]^B_{K_n} = (\vec{v}_1, \ldots, \vec{v}_n)$ and $D = diag_{m \times n}(\sigma_1, \ldots, \sigma_r)$ (K is the canonical basis). Also the dyadic decomposition is applicable, which states

$$A = \sigma_1 \vec{u}_1 \vec{v}_1^T + \ldots + \sigma_r \vec{u}_r \vec{v}_r^T$$

and is important, because it allows for best lower-rank approximations of the matrix A. The basis B contains vectors that maximize $||A\vec{x}||$. So vector $\vec{v}_1$ is of length one and depicts an element with the largest impact on $||A\vec{x}||$. Singular decomposition thus helps determine which elements affect $||A\vec{x}||$ and to what extent.

2.2 Energy Market and Volatility

Singular decomposition helps me eliminate market noise by approximating the matrix with its lower-rank representation. When considering the dataset as matrix A, each column of A is in the vector space of daily price changes. Therefore, multiplying the matrix A by a daily production profile vector $\vec{x}$ computes the historical daily financial profits and losses of that specific portfolio. The norm $||A\vec{x}||$ can be interpreted as the total historical volatility of a specific portfolio defined by the production vector $\vec{x}$. Before analyzing the vectors $(\vec{v}_1, \ldots, \vec{v}_n)$, singular values obtained by the SVD, shown in Table 3, allow to calculate the percentage of the total variation explained by each lower-rank approximation.

Table 3: Singular Values

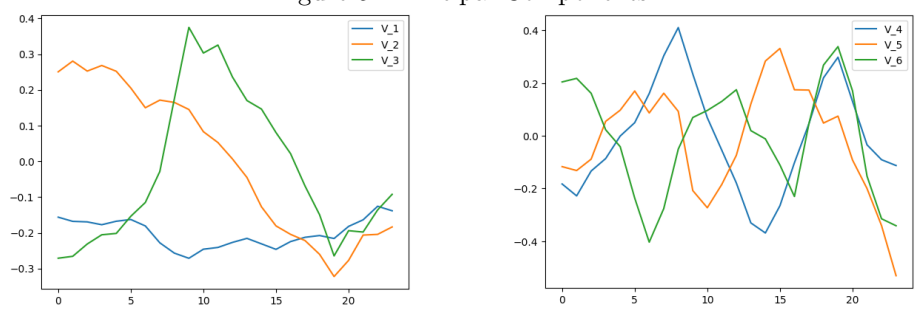
	Singular Value	R^2
σ_1	3486.7	0.603
σ_2	1641.5	0.737
σ_3	1215.1	0.810
σ_4	988.1	0.858
σ_5	762.2	0.887
σ_6	602.7	0.905

The second column of Table 3 is calculated via the R^2 formula

$$R^2 = \frac{ESS}{TSS} = \frac{\sum_{j=1}^{i} \sigma_j^2}{\sum_{t=1}^{24} \sigma_t^2},$$

where $j \leq i, i \in \{1, \ldots, 24\}$. According to these results, 90 % of total variance is explained by the first six columns of V. These vectors are often referred to as principal components in financial literature and they dictate the exact geometric shapes (seen in Figure 5) of independent market movements. So a

Figure 5: Principal Components



specific historical day in the dataset can be represented as a linear combination of these six curves with 90 % accuracy. By approximating the original matrix with a rank-six dyadic product, I get rid of unpredictable noise, which, if included, would lead to overfitting.

2.3 Numerical Mathematics

To further justify selection of the new lower-level rank, one can turn to numerical analysis. Calculating the relative error shows how much information is lost during data compression. Two common mathematical tools are used for this. The Euclidean norm measures the largest single error, so the relative error is denoted as

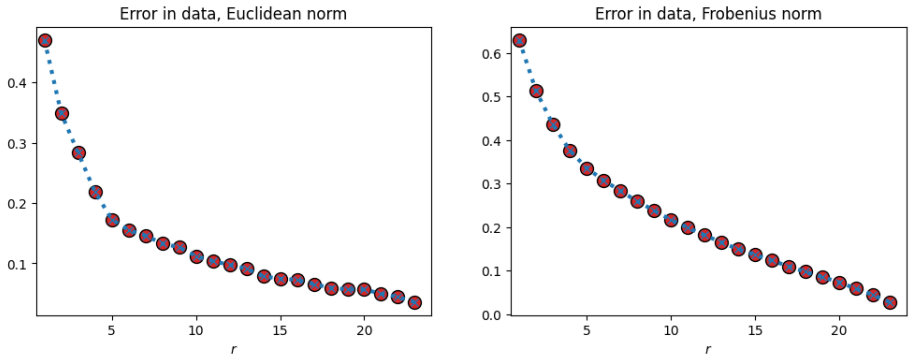
$$\frac{||A - A_r||_2}{||A||_2},$$

presented in red on the left of Figure 6. And the Frobenius norm measures the total size of all the dropped data,

$$\frac{||A - A_r||_F}{||A||_F},$$

also in red, seen on the right of the same figure. For both methods, the real error is calculated by comparing the new matrix to the original one.

Figure 6: Error and Rank relationship



The blue dashed line resembles the theoretical error, which comes only from the singular values. The theoretical Euclidean error is the first dropped singular value divided by the largest one,

$$\frac{\sigma_{r+1}}{\sigma_1},$$

and the theoretical Frobenius error adds up the squares of all the dropped singular values,

$$\sqrt{1-R^2} = \sqrt{\frac{\sigma_{r+1}^2 + \ldots + \sigma_{24}^2}{\sigma_1^2 + \ldots + \sigma_{24}^2}}.$$

This connects the error directly to the explained variance established earlier. The real and theoretical values visually match, which indicates the calculation is correct. The lines drop quickly at first and flatten out around a rank of six. Adding more components after this point does not improve accuracy much.

2.4 Economic Interpretation

To understand the geometric shapes of individual principal components further, I must further examine the French energy structure. From Figure 7 I am able to determine that most of French demand is satisfied by nuclear power plants and the rest is filled by renewable sources and fossil fuels.

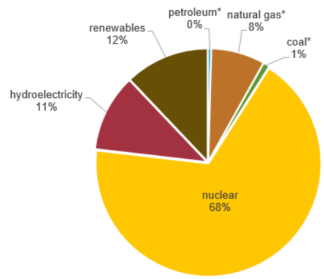


Figure 7: France Energy Structure

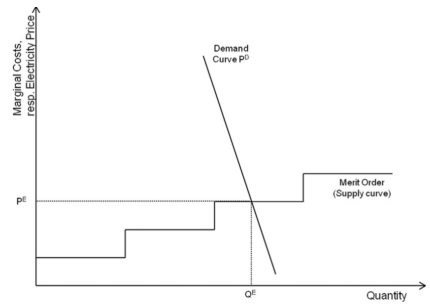


Figure 8: Supply and Demand

On the right side, Figure 8 shows an example of a possible supply and demand graph. The energy market utilizes a specific microeconomic model called the merit order curve. As there is an unsolved problem with storing energy, producers want to sell energy with the least marginal costs, making them cheaper. And so the supply curve is constructed by ordering all available power plants from the lowest marginal cost to the highest marginal cost. Nuclear power plants follow renewable resources and fossil fuel plants form the last section of the curve. Because most people do not check electricity price daily or in faster frequencies, making electricity consumption inelastic, the short-term demand curve is nearly vertical.

Generally known shifts in the demand curve are caused by the time of day. In the morning and evening, people and businesses consume more electricity, which shifts the vertical demand curve to the right. This means the demand curve intersects the steep fossil fuel section of the supply curve, causing higher prices. These daily demand cycles align with the first and second principal components, which represent the overall price shift and the morning to evening price spread.

On the supply side, solar power generation peaks at midday. Since solar energy has zero marginal costs, it pushes the supply curve to the right. Therefore, the demand curve crosses the supply curve at a much lower price point. This specific price drop is exactly what the third principal component captures. By linking these market mechanics to the linear algebra results, it is clear that the calculated vectors represent actual physical shifts in the grid.

3 Least Squares Hedging Optimization

Suppose owning a non-standard energy contract, that is a tailored agreement between a supplier and a customer. Let $\vec{x}$ be the production profile of this contract in one day. To hedge against this asset one must neutralize its risks.

3.1 Regression Model

So the first step towards achieving this goal is to find the profits and losses of $\vec{x}$ based on historical data. This can be achieved with a simple multiplication, $A\vec{x} = \vec{y}$. A linear model

$$\vec{y} = \beta_1 A\vec{v_1} + \ldots + \beta_6 A\vec{v_6} + \vec{\epsilon} \iff \vec{y} = X\vec{\beta} + \vec{\epsilon}$$

can be presented to find the effects of each principal component. There is no intercept included in this model, because a within transformation was used. Therefore if it was included it would be zero. Let V' be the first six columns of matrix V. With matrix algebra this model can be boiled down with the following steps

$$\vec{\beta} = ((AV')^T(AV'))^{-1}(AV')^T A\vec{x} =$$
$$= (I^T_{24\times 6} DU^T UDI_{24\times 6})^{-1}(I^T_{24\times 6} DU^T UDV^T)\vec{x} =$$
$$= D^{-2}_{6\times 6} D^2_{6\times 24} V^T \vec{x} = V'^T \vec{x},$$

algebraically represented as

$$x_i = \beta_1 v_{i,1} + \ldots + \beta_6 v_{i,6} + \epsilon_i.$$

The facts that $A = UDV^T$ and $U^T U = I$ were used in the calculations. This outcome brings an entirely new interpretation of the model, one that is not as intuitive. Because the financial profit is calculated by multiplying the hourly price by $\vec{x}$, finding the best hedge is the same as finding which principal components best overlap with the energy production. So regressing production vector $\vec{x}$ onto V' calculates those overlaps and measures how much of generation happens during the hours that an economic shock shifts the market.

3.2 Model Results

When dealing with a specific vector of production, for example functional values of the Gaussian bell $N(12, \sigma = 2.5)$ curve in an array, which are supposed to represent supply of solar energy, one may obtain regression results similar to the one presented in Table 4, where the explained variance $R^2 = 0.943$.

Table 4: Regression Results

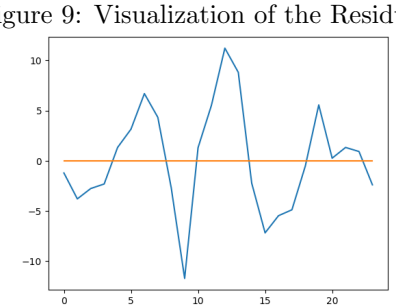
vector	coef	std err	t	$P > \|t\|$	[0.025	0.975]
$\vec{v}_1$	-73.3826	5.946	-12.341	0.000	-85.875	-60.890
$\vec{v}_2$	-3.0515	5.946	-0.513	0.614	-15.544	9.441
$\vec{v}_3$	63.0138	5.946	10.597	0.000	50.522	75.506
$\vec{v}_4$	-30.3312	5.946	-5.101	0.000	-42.824	-17.839
$\vec{v}_5$	3.4014	5.946	0.572	0.574	-9.091	15.894
$\vec{v}_6$	12.0263	5.946	2.023	0.058	-0.466	24.519

Since the coefficients have an unintuitive interpretation, it is easier to remember what happens when a linear regressing is run. A coefficient is the dot product of the dependent production vector and the respective principal component. The production vector has exposure to principal components one, three, four and six. Some of them are exposed in the positive direction and others in the negative. The coefficients of vectors two and five are statistically insignificant, which makes sense, since they are asymmetrical with positive and negative values and the vector $\vec{x}$ is symmetrical around noon, making the dot product close to zero. There is a small sample size of 24 and 18 degrees of freedom, thus any asymptotic properties can be ignored. The R^2 value is consistent with a statement made earlier, that a linear combination of the first six principal components can represent a day with 90% accuracy. Standard error is constant no matter the principal component, that can be explained by the formula used to calculate them, $\sigma^2(V^TV)^{-1}$, and orthonormality of V.

3.3 Residual & Assumptions Analysis

Figure 9 provides a visualization of the residuals.

Figure 9: Visualization of the Residuals



The graph displays a wave pattern instead of a random scatter of points, which indicates that the errors are autocorrelated. Also, the Durbin-Watson statistic is 1.017. This value being close to one hints to severe positive autocorrelation. Because the errors depend on previous hours, the standard regression assumption of independent residuals is violated. The variance of the residuals is not constant across the day, as the solar profile produces zero electricity at night and reaches its peak at noon. The errors are completely flat at night and swing heavily during the day, which indicated heteroskedasticity. The residuals still keep a perfectly symmetrical shape over the day. The Jarque-Bera test gives a probability value of 0.991, indicating the residuals follow a normal distribution. Since the goal of this model is to find the overlap for hedging rather than hypothesis testing, the model remains usable for financial applications.

3.4 Hedging Strategy

The regression results state the exact market exposure of the tailored energy asset. To stabilize revenues, one must neutralize these risks by taking opposite financial positions of standard contracts.

Standard market products usually include base contracts covering the entire day. The principal component exposure of these standard contracts needs to be calculated. Comparing the standard contract exposures to the tailored asset creates a system of equations and solving this system provides the exact volume of standard contracts required to sell on the exchange. This resulting portfolio offsets the hidden risk factors.

4 Conclusion

This project applied SVD to historical day-ahead electricity prices to find an optimal hedging curve. By transforming the data into stationary daily price differences, the principal components were extracted. Numerical analysis demonstrated that a small number of principal components can explain 90 % of market volatility. Furthermore, the principal components mirror economic movements of the supply shifts and the merit order curve. Regressing a production profile onto these components reveals the vulnerability of the non-standard asset to independent market shocks.